Vidyalankar Institute of Technology
Accredited A+ by NAAC

Version 09/27-09

Consolidated Academic Administration Plan for the Course PCIT12T Software Engineering with WDL (core) Sem. V – Program Information Technology 2026-27 Odd Semester Faculty - Prof. Deepali Shrikhande (Cluster Mentor)

The academic resources available in VIT –

VMIS (ERP)	V-Refer and V-Live	VIT Library	VAC & MOOC Courses
Institute & Department Vision and Mission	Former IA question papers and solutions (prepared by faculty)	Former IA question papers solutions - hardcopy	Value Added Courses (VAC) are conducted throughout the semester & in the semester break - Enrol for the VACs
Program Educational Objectives (PEO)	MU end semester examination question papers and solutions (prepared by faculty)	MU end semester exam question paper & solutions - by faculty, hardcopy	
Program Specific Outcome (PSO)	Class notes and Digital Content for the subject (scanned / typed by faculty)	All text books, reference books, e -books mentioned in the syllabus & AAP	Online courses from NPTEL, Coursera etc. are pursued throughout the semester - Register for the course & get certified
Program Outcome (PO)	Comprehensive question bank, EQ, GQ, PPT, Class Test papers	Technical journals and magazines for reference	
Departmental Knowledge Map	Academic Administration Plan & Beyond Syllabus Activity report	VIT library has many resources e:g :- IEEE, Nimbus, xplore, EBSCO etc.	Watch former lectures captured in LMS at VIT

1.a Course Objectives (Write in detail – as per NBA guidelines)

Cognitive	What do you want students to know?	To understand Concepts, processes and standards used in software development
Affective	What do you want students to think / care about?	To think about the proper process model to be applied and create user friendly software
Behavioural	What do you want students to be able to do?	To design, test and implement software as per user requirements

Advice to Students:

Attend every class!!! Missing even one class can have a substantial effect on your ability to understand the course. Be prepared to think and concentrate, in the class and outside. I will try to make the class very interactive. Participate in the class discussions. Ask questions when you don't understand something. Keep up with the class readings. Start assignments and homework early. Meet me in office hour to discuss ideas, solutions or to check if, what you understand is correct.

The v-Refer Link :

https://vrefer.vidyalankar.edu.in/index.php/apps/files/favorites/920858?dir=/vit%20vrefer/INFT/SEM%20V/2026-27/Verticals/PC_PCC/SEWDL%20%28PCIT12T%26P%29

Creation of microsite or teams link :

DIV-A- General | Odd2627-TYINFT-A-SEWDL-TH-DSJ

DIV-B- General | Odd2627-TYINFT-B-SEWDL-TH-DSJ

DIV C - General | Odd2627-TYINFT-C-SEWDL-TH-DSJ

Collaboration Policy:

We encourage discussion between students regarding the course material. However, no discussion of any sort is allowed with anyone on the assignment and homework for the class. If you find solution to some problems in a book or on the internet, you may use their idea for the solution; provided you acknowledge the source (name and page in the book or the website, if the idea is found on the internet). Even though you are allowed to use ideas from another source, you must write the solution in your own words. If you are unsure whether or not certain kinds of collaboration is possible, please ask the teacher.

1.b Course Outcome (CO) Statements and Module-Wise Mapping (follow NBA guideline)

CO No.	Statements	Related Module/s
CO1	**Remember** fundamental concepts of web technologies , and foundational software development life cycle (SDLC) models such as Waterfall, Incremental, and Spiral.	2
CO2	**Understand** the software requirement engineering process, including requirement types, elicitation techniques, and agile methodologies like Scrum and Extreme Programming.	2,3
CO3	**Apply** HTML5, CSS3, and JavaScript to develop user-centric web interfaces following established interaction design heuristics and frontend development best practices.	1,3
CO4	**Analyse** software process models, estimation techniques such as COCOMO, and project metrics to determine suitable approaches for diverse software project scenarios.	4
CO5	**Evaluate** software quality assurance strategies including testing techniques, risk mitigation plans (RMMM), and configuration management processes.	6
CO6	**Create** a responsive web application using HTML, CSS, and JavaScript, integrating principles of project scheduling, task decomposition (WBS), and version control systems.	1,4,5

1.c Course Outcome (CO) Statements mapping with Learning Outcome (LO)

	Mapped to Learning Outcomes
CO1	1.1, 1.2, 2.1, 2.3
CO2	2.2, 2.3, 3.1, 3.2, 3.3, 5.1
CO3	1.1, 1.2, 1.3, 1.4, 5.2, 5.3
CO4	2.2, 2.4, 4.1, 4.2, 5.1
CO5	6.1, 6.2, 6.3
CO6	1.3, 1.4, 4.3, 4.4

1.d Mapping of COs with POs (As per new detailed Syllabus)

(List of POs is available in V-refer)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
	Knowledge A	Analysis	Design	Investigation	Modern Tools	Society	Environment &sustainability	Ethics T	Teamwork C	Communication	Project Mgt	Lifelong learning
CO 1	S					-	S		W	W		S
CO 2	W		W	S		-	-		W	S		S
CO 3	W	S	S			-	-		S	W		
CO 4	M	W		W	S	-	-	W			S	
CO 5	W			M	W	-	-	W			W	S
CO6	W				W	S	-	S				W

1.e Mapping of COs with PSOs (As per new detailed Syllabus)

	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	S	M	M	M
CO 2	W	W	W	W
CO 3	M	M	M	M
CO 4	S	M	W	W
CO 5	M	W	M	M
CO6	S	M	M	M

1.f Teaching and Examination Scheme (As specified by the autonomous syllabus) for the Course

Verticals	BSC/ESC	Program Courses	Multidisciplinary Courses	Skill Courses	HSSM	Experiential Learning	Liberal Learning
Tick suitable category		√					

Subject Code	Subject Name	Teaching Scheme			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/Practical	Tutorial	Total
PCIT12T	Software Engineering with WDL	2	-	-	2	-	-	3

Subject Code	Subject Name	Examination Scheme							
		Theory			Total (Theory)	Practical			Total (Practical)
		ISA	MSE	ESE		ISA	ESE	ORAL	
PCIT12T	Software Engineering with WDL	15	40		75				-

Subject Code	Subject Name	MSE-1*		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
IT12T	Software Engineering with WDL	1	50% from Module2 based on assignments with slightly enhance difficulty /complex 25% from Module 3 based on assignments with slightly enhance difficulty /complex	L1, L2, L3, L4
IT12T	Software Engineering with WDL	2	25% from Module 2 based on Assignments 25% from Module 1 based on Assignments	L2, L3, L4, L5

Subject Code	Subject Name	MSE-2*		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
IT12T	Software Engineering with WDL	1	50% from Module5 based on assignments with slightly enhance difficulty /complex 25% from Module 6 based on assignments with slightly enhance difficulty /complex	L2, L3 ,L4 ,L5
IT12T	Software Engineering with WDL	2	25% from Module 5 based on Assignments 25% from Module 4 based on Assignments	L2, L3 ,L4

Subject Code	Subject Name	ESE#		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
IT12T	Software Engineering with WDL	1	2 Questions 05 Marks each from Assignments	L1, L2, L3, L4
IT12T	Software Engineering with WDL	2	2 Questions 05 Marks each from Assignments	L2, L3 ,L4
IT12T	Software Engineering with WDL	3	1 Questions 10 Marks from Assignments	L1, L2, L3, L4
IT12T	Software Engineering with WDL	4	1 Questions 10 Marks from Assignments	L2, L3 ,L4

*** Recommended distribution: -** 30 Marks from Assignments, 40 marks based on assignments with slightly enhance difficulty /complex, 30 marks from thought provoking

Recommended distribution: - 30 Marks from Assignments, 40 marks based on assignments/MSE with slightly enhance difficulty /complex, 30 marks from thought provoking

1.g Faculty-Wise Distribution of all Lecture-Practical-Tutorial Hours for the Course

Divisions	Lecture (Hrs.)	Practical (Hrs.)				Tutorial (Hrs.)			
		Batch 1	Batch 2	Batch 3	Batch 4	Batch 1	Batch 2	Batch 3	Batch 4
A	DSJ	DSJ	DJS	DG	DSJ				
B	DSJ	DSJ	DJS	DG	DSJ				
C	DSJ	DG	DG	DG	DSJ				

1.h Office Hours (Faculty will be available in office in this duration for solving students' query)

Division	Day	Time (at least 1 Hr. / Division)	Venue (Office Room No.)
A	Wednesday	4 PM – 5 PM	BS E105
B	Wednesday	4 PM – 5 PM	DSJ E105
C	Wednesday	4 PM – 5 PM	DSJ E105

Note: Office hours should be after academic hours of students.

2.a Syllabus: Module Wise Teaching Hours and % Weightage in autonomous syllabus Question Paper

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
1	Introduction to Web Programming and Concepts Introduction to HTML, HTML Document Structure ,Text Elements, Images and Attributes, Hyperlinks, Semantic HTML, complex image maps, tables and nested tables, Inserting web page, Setting & modifying field properties, Validating HTML CSS: Internal and External CSS, CSS Grid Overview, Sizing Grid Columns and Rows, Building a Simple CSS Grid Layout JavaScript & Document Object Model: Introduction to JavaScript, Variables and Objects, Decision Making Statement, Loops, Arrays, Functions & Prototypes, Core JavaScript Objects, DOM Introduction, Event Model, Function	08	10	25	25	PI 1.1.6, PI 1.3.3 PI 2.1.2, PI 2.1.3 PI 5.1.3, PI 5.2.3, PI 3.2.3 ,PI 3.3.4
Learning Outcome-1.1	Remember the structure and syntax of an HTML document including DOCTYPE.					
Learning Outcome-1.2	Understand the role of text formatting tags, image embedding with attributes					
Learning Outcome-1.3	Understand and create grid-based CSS layouts.					
Learning Outcome-1.4	Develop dynamic and interactive web components using JavaScript and DOM concepts					
Learning Outcome-1.5	Debug and resolve issues in front-end code implementations.					
2	The Software Process Generic view of Process, Prescriptive Models: Waterfall Model, Incremental-RAD Model, Evolutionary Process Model-Prototyping, Spiral Agile Methodology, Scrum and Extreme Programming	04	20	50	50	PI 2.1.2, PI 2.1.4 PI 3.1.4, PI 2.2.4, PI 3.1.2, PI 3.3.4
Learning Outcome-2.1	Understand the structure of the software engineering process.					
Learning Outcome-2.2	Learn to analyze project requirements and choose an appropriate software engineering model based on project scope, complexity, and constraints.					
Learning Outcome-2.3	Understand the principles and concepts underlying different software engineering models.					
Learning Outcome-2.4	Compare traditional and agile software models to evaluate their suitability for specific project needs.					
3	Requirements Engineering and Analysis Requirement, Types of Requirements, Requirement Gathering, Requirement	04	20	25	25	PI 2.1.2, PI 3.1.2, PI 2.2.2, PI 2.2.4, PI 3.1.2, PI 3.2.2

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
	Engineering Task, SRS (Software Requirement Specification)					
Learning Outcome-3.1	Understand the importance and classification of software requirements, including functional and non-functional types.					
Learning Outcome-3.2	Describe and compare various requirement gathering techniques such as interviews, questionnaires, and observations.					
Learning Outcome-3.3	Explain the components and significance of a Software Requirement Specification (SRS) document and its role in the requirements engineering process.					
4	Software Estimation and Scheduling Management Spectrum, 4Ps (people, product and process), Process and Project metrics, Software Project Estimation: LOC, FP, Empirical Estimation Models - COCOMO Model, Project scheduling: WBS, defining a Task Set for the Software Project, Timeline charts, Tracking the Schedule	05	10	25	25	P.I. – 1.4.1, 4.3.1, 11.2.1, 5.2.1
Learning Outcome-4.1	Describe the elements of the management spectrum and analyze the impact of the 4Ps on software project success.					
Learning Outcome-4.2	Apply estimation techniques like LOC, Function Points, and the COCOMO model to estimate project effort and cost.					
Learning Outcome-4.3	Develop a project schedule using WBS, timeline charts, and apply methods to monitor and track project progress.					
Learning Outcome-4.4	Interpret and create basic timeline charts (like Gantt charts) to visualize project schedules and dependencies based on task set for a software project based on scope, milestones, and deliverables.					
5	Design Engineering Software Design Concepts, Interaction Design, Golden Rules and Heuristics.	03	10	50	50	1.4.1, 3.4.1, 3.2.1, 5.1.1
Learning Outcome-5.1	Understand core software design concepts such as modularity, abstraction, and separation of concerns.					
Learning Outcome-5.2	Explain the principles of interaction design and their role in user experience.					
Learning Outcome-5.3	Apply design heuristics and golden rules to evaluate and improve user interface quality.					
6	Software Testing and Risk Management Testing: Software Quality, Testing: Strategic Approach, Strategic Issues- Testing: Strategies for Conventional Software. Risk Management: Risk Identification, Risk Assessment, Risk Projection, RMMM, Software Configuration management, SCM process- Version Control, Change Control	06	30	25	25	PI 6.2.3, PI 4.1.1, PI 7.1.1, PI 4.1.3, PI 5.2.1, PI 5.3.4

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
Learning Outcome-6.1	Define software quality attributes and describe strategic approaches to software testing.					
Learning Outcome-6.2	Explain risk management activities including identification, assessment, projection, and mitigation using RMMM. (CO5)					
Learning Outcome-6.3	Apply software testing strategies to conventional software systems to ensure reliability.					
Learning Outcome-6.4	Describe the software configuration management process and use version and change control practices.					
* Insert rows for more modules in the Course Total		30	100	100	100	

Note: - As an attachment Annexure is required for assessment criteria of learning outcomes.

2.b Prerequisite Courses

No.	Semester	Name of the Course	Topic/s
1	-	NIL	-

2.c Relevance to Future Courses

No.	Semester	Name of the Course
1	VI & VII	Final Year Project
2	VI	STQA

2.d See :- Identify real life scenarios/examples which uses the knowledge of the subject ,(Discussion on how to prepare examples and case studies e.g. "Boeing Plane": C Programming Language – Intro to Computer Science – Harvard's CS50 (2018) – Bing video)

Real Life Scenario	Concept Used
Payroll management system	Analysis, Design, implementation, Testing and maintenance
Online shopping system	Analysis, Design, implementation, Testing and maintenance
Online banking	Analysis, Design, implementation, Testing and maintenance

3 Past Results – Division-Wise

Details	Target – DEC 2026	DEC 2025	DEC 2024	MAY 2024
Course Passing % – Average of 2 Divisions	100%	100%	100%	100%
Marks Obtained by Course Topper (mark/100)	75	73/75	75	71 / 75

	Division A		Division B		Division C	
Year	Initials of Teacher	% Result	Initials of Teacher	% Result	Initials of Teacher	% Result
DEC 2025	DST	100%	DSJ	100%	DSJ	100%
DEC 2024	DSJ	100%	DSJ	100%	NA	NA
DEC 2023	NA	NA	NA	NA	NA	NA

4 All the Learning Resources – Books and E-Resources

4.a List of Textbooks (T – Symbol for Textbooks) to be Referred by Students

Sr. No	Textbook Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Software Engineering: A Practitioner 's Approach	Roger Pressman	McGraw-Hill Publications	-	1,2,3,4,5,6	Yes
2	Software Engineering	Ian Sommerville,	Pearson Education	(9th edition)	2,4	Yes
3	Software Engineering Fundamentals	Ali Behfrooz and Fredeick J.Hudson	Oxford University Press	-	1,2,3,4,5,6	Yes
4	HTML 5 Black Book (Covers CSS3, JavaScript, XML, XHTML, AJAX, PHP, jQuerY)		DT Editorial Services	2ND Edition	1	Yes

4.b List of Reference Books (R – Symbol for Reference Books) to be Referred by Students

Sr. No	Reference Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Software Engineering – Concepts and Practices	Ugrasen Suman	Cengage Learning	-	1,2,3,4,5,6	Yes
2	An integrated approach to Software Engineering	Pankaj Jalote,	Springer/ Narosa	-	1,2,3,4,5,6	Yes
3	Software Engineering	Jibitesh Mishra and Ashok Mohanty,	Pearson	-	1,2,3,4,5,6	Yes
4	Fundamentals of Software Engineering	Rajib Mall	Prentice Hall India	-	1,2,3,4,5,6	Yes

4.c List of E - Books (E – Symbol for E-Books) to be Referred by Students

Sr. No	E- Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Software Engineering: A Practitioner's Approach	Roger Pressman	McGraw-Hill Publications	7th	1,2,3,4,5,6	Yes
2	The Complete Reference – HTML CSS	Thomas Powell	McGraw-Hill Publications	5th	1	Yes

4.d Reading latest / top rated research papers (at least 5 papers)

Name of Paper	Name of Authors (Background)	Published in		Problem Statement	Available in our Library
		Date	Journal		
Demystifying LLM-Based Software Engineering Agents	**Chunqiu Steven Xia, Yinlin Deng, Soren Dunn and Lingming Zhang.**	JULY 2025	Proceedings of the ACM on Software Engineering, Vol. 2, FSE, Article FSE037	Autonomous software-engineering agents are complex, expensive and difficult to debug because they perform long sequences of tool-based actions. The paper investigates whether a simpler agentless process involving fault localisation, repair and patch validation can solve software issues effectively.	YES
On the Effectiveness of LLM-as-a-Judge for Code Generation and Summarization	**Giuseppe Crupi, Rosalia Tufano, Alejandro Velasco, Antonio Mastropaolo, Denys Poshyvanyk and Gabriele Bavota.**	August 2025	IEEE Transactions on Software Engineering, Vol. 51, No. 8, pp. 2329–2345,	Human evaluation of generated code and code summaries is costly, while similarity metrics such as BLEU may not accurately measure correctness or quality. The study evaluates whether LLMs can reliably judge generated programs and code summaries instead of human evaluators.	YES
ROSEAU: Fast, Accurate, Source-Based API Breaking Change Analysis in Java	**Corentin Latappy, Thomas Degueule, Jean-Rémy Falleri, Romain Robbes and Lina Ochoa.**	SEPT 2025	41st IEEE International Conference on Software Maintenance and Evolution — ICSME 2025, pp. 517–528,	Existing Java API compatibility tools mainly depend on compiled JAR files, limiting commit-level analysis and large-scale studies of API evolution. The paper develops ROSEAU to detect breaking API changes directly from source code or bytecode quickly and accurately.	YES
RepoScope: Leveraging Call	**Yang Liu, Li Zhang, Fang Liu,**	APRIL 2026	48th IEEE/ACM International	Repository-level code generation systems often retrieve code that	YES

Chain-Aware Multi-View Context for Repository-Level Code Generation	**Zhuohang Wang, Donglin Wei, Zhishuo Yang, Kechi Zhang, Jia Li and Lin Shi.**		Conference on Software Engineering — ICSE 2026,	is textually similar but structurally irrelevant and fail to capture dependencies between functions. RepoScope constructs a repository semantic graph and retrieves caller, callee and similarity-based contexts to improve generated code.	
CASCADE: LLM-Powered JavaScript Deobfuscator at Google	**Shan Jiang, Pranoy Kovuri, David Tao and Zhixun Tan**	APRIL 2026	48th IEEE/ACM International Conference on Software Engineering—Software Engineering in Practice — ICSE-SEIP 2026,	JavaScript obfuscation makes programs difficult to understand, test and analyse and creates problems for malware detection. CASCADE combines Gemini with a compiler intermediate representation to recover strings, API names and program behaviour from heavily obfuscated JavaScript.	YES

4.e **Based on research paper an identify the current Problem statement**

Problem Statement			**Used in**			
	Quiz	Assignment	*Mini Project*	Poster Presentation	Test	Any Other
Autonomous software-engineering agents are complex, expensive and difficult to debug because they perform long sequences of tool-based actions. The paper investigates whether a simpler agentless process involving fault localisation, repair and patch validation can solve software issues effectively.				√		
Human evaluation of generated code and code summaries is costly, while similarity metrics such as BLEU may not accurately measure correctness or quality. The study evaluates whether LLMs can reliably judge generated programs and code summaries instead of human evaluators.				√		
Existing Java API compatibility tools mainly depend on compiled JAR files, limiting commit-level analysis and large-scale studies of API evolution. The paper develops ROSEAU to detect breaking API changes directly from source code or bytecode quickly and accurately.				√		
Repository-level code generation systems often retrieve code that is textually similar but structurally irrelevant and fail to capture dependencies between functions. RepoScope constructs a repository semantic				√		

graph and retrieves caller, callee and similarity-based contexts to improve generated code.						
JavaScript obfuscation makes programs difficult to understand, test and analyse and creates problems for malware detection. CASCADE combines Gemini with a compiler intermediate representation to recover strings, API names and program behaviour from heavily obfuscated JavaScript.				√		

4.f Identify Companies / Industries which use the knowledge of the subject and thus may provide Internships and final Placements

Name of the Company	To be / Contacted for		
	Student Internship	Student Final Placement	Faculty Internship
TCS	√	√	
Google	√	√	
Microsoft	√	√	

4.g Identify suitable relevant TOP Guest Speakers from Industry, Example: - (CS50 Lecture by Mark Zuckerberg - 7 December 2005 - YouTube)

Name of the Identified Guest Speaker	Designation	Name of the Company
Dr. Ninad Chaudhary	Instructor	Albany University
Mr.Yash Trivedi	Program Manager	
Mr. Anand Kelkar	Sr. Tech Program Manager	Amazon

4.h Identify relevant technical competitions to participate [Competitions -Paper Presentations, Projects, Hackathons, IVs etc..]

Name of the Relevant Technical Competition Identified to participate	Organized by	Date of the Event
SIH	AICTE	As per announcement
ICSED 2026	ACM	October 29-31, 2026

4.i Identify faculty in TOP schools / Universities who are teaching same / similar subject and develop rapport e.g. Exchange Lecture Material (Assignments / Tests / Project etc..), Joint Paper Publication

University	Name of the Course	Name of Faculty	Type of Collaboration		
			Exchange of Lecture Material	Joint Publication/ Research	Other
IIT Kanpur	Software Engineering	Mr.Pankaj Jalote	https://onlinecourses.nptel.ac.in/noc24_cs127/preview	-	Book reference
Albany University	Software Engineering	Dr. Ninad Chaudhary	Exchange of Lecture Material		

4.j Module Best Available in – Title of the best resource [from *4.a* to *4.d* in this AAP] and other details as necessary

Module No.	Title of the Module	Textbook	Mention the Tile					
			Reference Book	E-books	Journal	E-Journal	Available in our Library	V-refer
1	Introduction to Web Programming and Concepts	T4					Yes	
2	The Software Process	T1, T2					Yes	
3	Requirements Engineering and Analysis	T1					Yes	
4	Software Estimation and Scheduling	T1					Yes	
5	Design Engineering	T1					Yes	
6	Software Testing and Risk Management	T1					Yes	

4.k Referred to any top-rated university in that subject for content

University	Name of the Course	Name of Faculty	Date of Delivery of the Course	Remarks
Massachusetts Institute of Technology (MIT)	Software Engineering	Prof. Nancy Leveson	Fall 2005	https://ocw.mit.edu/courses/ 16-355j-software- engineering-concepts-fall-2005/pages/lecture-notes/
University of Virginia	Agile meets design thinking	Alex Cowan	Self Paced	https://www.coursera.org/learn/uva-darden-getting- started-agile
NPTEL (IIT Madras)	Software Engineering	Prof. Rajib Mall (IIT KGP)	July–Dec 2018	https://nptel.ac.in/courses/10 6105087

4.l Faculty received any certification related to this subject. List of Certifications Identified / Done

Course	Certifying Agency	No. of Hours	Level of the Course		Certification		Remarks
			Introductory	Advance Skill Development	Done on	Proposed to be on	
Software Processes and Agile Practices	Coursera	12	Yes		May 2020		-
Algorithms and Software Engineering for Professionals	Eduonix	10	Yes		May 2022		
The complete software testing boothcamp	Udemy	30.5		Yes	Dec 2023		

4.m Completed subject wise/cluster wise training with cluster mentor. List of relevant Refresher Course Identified / Done

Course	Certifying Agency (As suggested by DAB/Cluster Mentor/Industry/University other than MU)	Certification		Remarks
		Done on	Proposed to be on	
PBL				
Sub. Content Training				

4.n Best Practices Identified and adopted

No.	Item	Best Practices Identified		
		IIT Kharagpur Dr. Debasis Samanta	IIT Kharagpur Prof. Rajib Mall	Massachusetts Institute of Technology Prof. Nancy Leveson
1	Microsite	Yes – Course website with lecture slides and structure	Yes – NPTEL course microsite	Yes – OCW course page with structure
2	Video Lectures		Yes	
3	Assignments	Yes	Yes	Yes
4	Mini Project			Yes
5	Assessment Metric			
6	Quizzes		Yes	Yes
7	Labs/ Practical (PBL)			
8	Tests			
9	Peer Assessment			
10	Any Other	Presentation-based delivery with self-study emphasis	YouTube availability, recognized NPTEL certification	Publicly licensed OCW content for reuse

4.o Web Links for Online Notes/YouTube/VIT Digital Content/VIT Lecture Capture/NPTEL Videos

Students can view lectures by VIT professors, captured through LMS 'Lecture Capture' in VIT campus for previous years.

No.	Websites / Links	Module Nos.
1	http://freevideolectures.com/Course/2318/Software-Engineering	2,3,4,5,6
2	https://freevideolectures.com/subject/web-designing/	1

4.p Recommended MOOC Courses like Coursera / NPTEL / MIT-OCW / edX/VAC etc.

Sr. No.	MOOC Course Link	Course conducted by – Person / University / Institute / Industry	Course Duration	Certificate (Y / N)
1	ChatGPT for Beginners https://www.mygreatlearning.com/academy/learn-for-free/courses/chatgpt-for-beginners	Great Learning	2	Y

5 Consolidated Course Lesson Plan

	From (date/month/year)	From (date/month/year)	Total Number of Weeks
Semester Duration	06/07/2026	16/10/2026	13

Week	Lecture no.	Module No.	Lecture Topics / IA 1 and IA 2 / BSA planned to be covered	Actual date of Completion (Hand written)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
1	1 2	2	Generic view of Process, Prescriptive Models: Waterfall Model, Incremental-RAD Model, Evolutionary Process Model-Prototyping, Spiral		5 6			T4 / CH2/31
2	3 4	2	Agile Methodology, Scrum and Extreme Programming		5 6			T4 / CH2/31
3	5 6	3	Requirement, Types of Requirements, Requirement Engineering Task,		5 6			T4 / CH18 /465
4	7 8	3	Requirement Gathering , SRS (Software Requirement Specification)		5 6			T4 / part 4/265
5	9 10	4	Management Spectrum, 4Ps (people, product and process) ,Process and Project metrics, Software Project Estimation: LOC, FP,		1 3			T1/ch- 1/3
6	11 12	4	Empirical Estimation Models - COCOMO Model,		1 3			T1/ch-3/65
7	13-14	4	Project scheduling: WBS, Defining a Task Set for the Software Project, Timeline charts, Tracking the Schedule		2			T1/ch-5/128
8	15-16	5	Software Design Concepts, Interaction Design , Design Golden Rules and Heuristics.		2			T1/ch-5/128
9	17-18	6	Testing: Software Quality, Testing: Strategic Approach, Strategic Issues-Testing: Strategies for Conventional Software.		3			T1/ch-26/692
10	19-20	6	Risk Management: Risk Identification, Risk Assessment, Risk Projection, RMMM, Software		3			T1/ch-27/721

			Configuration management,					
11	21-22	6	SCM process- Version Control , Change Control		3			T1/ch-27/721
12	2324	1	Introduction to HTML, HTML Document Structure Text Elements, Images and Attributes, Hyperlinks, Semantic HTML,		6			T1/ch-9/242
13	2526	1	complex image maps, tables and nested tables, Inserting web page, Setting & modifying field properties, Validating HTML		4			T1/ch-17/449
14	2728	1	CSS: Internal and External CSS, CSS Grid Overview, Sizing Grid Columns and Rows, Building a Simple CSS Grid Layout		4			T1/ch-28/744
15	29-30	1	Javascript & Document Object Model: Introduction to JavaScript, Variables and Objects, Decision Making Statement, Loops, Arrays, Functions & Prototypes, Core JavaScript Objects, DOM Introduction, Event Model, Function		4			T1/ch-22/584

6 Rubric for Grading and Marking of Term Work (inform students at the beginning of semester)

- Activity/ies should be designed as per reference of credit structure.
- If the subject is of 2 credit, activity/ assignment should be design for 2 hours with appropriate complexity and engaging time.

Theory (ISA=15)												Total
Class Participation	Assignment -1	Assignment -2	Assignment -3	Assignment -4	Assignment -5	Assignment -6	Assignment -7	Assignment -8	Assignment -9	Assignment -10	Assignment -11	
Pop quiz- 2 VIDEO SEMINAR - 2 THT -1	1	1	1	1	1	1	1	1	1	1	1	15

Class Participation	Practical -1	Practical -2	Practical -3	Practical -4	Practical -5	Practical -6	Practical -7	Practical -8	Practical -9	Practical -10	TOTAL
Mini project & PRESENTATION – 10 Mini Project Paper - 6	1	1	1	1	1	1	1	1	1	1	25

Class Participation	MSE-1	MSE-2	ESE	Total
As per attendance and activity participation	20	20	40	75

7 Assignments / Tutorials Details

Assignment/ Tutorial No.	Title of the Assignments / Tutorials	CO Map	Mapping Bloom Taxonomy Level	Assignment/ Tutorials given to Students on	Assignments to be submitted back on
1	Assignment 1- SDLC MODELS	2		Week 1	Week 2
2	Assignment 2- AGILE MODELS	2		Week 2	Week 3
3	Assignment 3- IEEE PAPER REVIEW AND POSTER PRESENTATION	1-6		Week 3	Week 4
4	Assignment 4- REQUIREMENT ENGINEERING	3		Week 4	Week 6
5	Assignment 5- SOFTWARE MANAGEMENT	1		Week 5	Week 5
6	Assignment 6- PROJECT SCHEDULING	4		Week 6	Week 8
7	Assignment 7- DESIGN CONCEPTS	5		Week 7	Week 8
8	Assignment 8- UML	6		Week 8	Week 9
9	Assignment 9- TESTING	6		Week 9	Week 10
10	Assignment 10- RISK MANAGEMENT	5		Week 10	Week 11
11	Assignment 11- SOFTWARE CONFIGRATION MANAGEMENT	6		Week 11	Week 12

Analysis of Assignment / Tutorial Questions and Related Resources

Assignment / Tutorial	Week No.	Type* (√)			Module No.	Based on #			Question Type (√)	
		OT	CS	DTP		Textbook	Reference Book	Other Learning Resource	Real Life Assignments	Thought Provoking
1	Week 1	√		√	2			√	√	√
2	Week 2	√		√	2			√	√	√
3	Week 3	√		√	1-6			√	√	√
4	Week 4	√			3			√	√	
5	Week 5	√			1			√	√	
6	Week 6	√		√	4			√	√	√
7	Week 7	√		√	5			√	√	√
8	Week 8	√		√	6			√	√	√
9	Week 9	√			6			√	√	
10	Week 10	√			5			√	√	
11	Week 11	√		√	6			√	√	√

* Tick (√) the Type of the Assignment: Online Tools (OT); Collaborative Assignments (CS); Design /Thought provoking (DTP)
Write number for textbook, reference book, other learning resource from this AAP – *from Points 4.a to 4.d*

8 In Semester Assessment (ISE) / Other Class Test / Open Book Test (OBT)/Take Home Test (THT) Details

Tests	Test Dates	Module No.	CO Map	MSE Question Paper Pattern	Policy
ISE		1-6	CO1-CO6		As per autonomous assessment policy
Pop Quiz		1-6	CO1-CO6		Quiz for revision and concept checking
Open Book Test	NA	NA	NA	NA	NA
Take Home Test	YES	4	CO4		Assignment-based evaluation
Class tests / prelims	NA	NA	NA	NA	NA
Class tests / prelims	NA	NA	NA	NA	NA
Any other test/exams	NA	NA	NA	NA	NA

*** Failures of IA test (IA1+IA2) shall appear for IA test in the next semester. There is no provision for re-test in the same semester.**

9. Practical Activities

Practical No.	Module No.	Title of the Experiments	Type of Experiment		Topics to be highlighted	CO Map
			PBL	Newly Added		
1	2	Project Selection and Conceptualization			Project Selection	CO1
2	3	Prepare SRS for the Project topic (Private study/Home work)			SRS	CO2
3	1	Create Simple web page using HTML5			HTML	CO5
4	1	Design a web page using CSS (Cascading Style Sheets) which includes the following: Use different font, styles: In the style definition you define how each selector should work (font, color etc.). Then, in the body of your pages, you refer to these selectors to activate the styles.	YES		CSS	CO6
5	3	PBL: For your Project Scenario: A user attempts to submit a registration form with an invalid email and a short password. Write JavaScript to validate the following fields of the Registration page. First Name (Name should contain alphabets and the length should not be less than 6 characters). Password (Password should not be less than 6 characters in length). E-mail id (should not contain any invalid characters and must follow the standard pattern name@domain.com) Mobile Number (Phone number should contain 10 digits only). Last Name and Address (should not be Empty).	Yes		Java Script	CO6
6	5	Prepare DFD-Data flow diagram for the Project topic			DFD	CO3
7	5	Prepare Use case diagram for the Project topic Prepare Sequence Activity diagram for the Project topic			UML DIAGRAMS	CO3
8	5	Prepare Component and Deployment diagram for the Project topic			UML DIAGRAMS	CO3
9	4	Prepare WBS and Gantt Chart for the Project topic	YES		WBS	CO3
10	6	Prepare Test Case plan for Project topic			TESTING	CO4
11	6	Prepare RMMM Document for Project topic			RISK	CO4
12	1-6	MINI PROJECT				

10 Uncovering syllabus with different Activities.

No.	Type of the Activity	Activities	Number of beneficiaries	Other Details – guest profile, feedback, mark sheet, report
1	Experiential learning/Interaction with Outside World	1- Guest Lectures by Industry Expert		Guest Session by - Mr. Kartik Kansaria
		2- Workshops		
		3- Mini Project		In WDL lab
		4- Industrial Visit		
		5- Any other activity		
2	Collaborative & Group Activity	6- Poster Presentation		Poster in a group 2 students
		7- Minute Papers		
		8- Students Seminars		In group of 2
		9- Students Debates		
		10- Panel Discussion / Mock GD		
		11- Mock Interview		
		12- Any other activity		Role Play on Project team management
3	Co-Curricular Activity	13- Informative videos (NPTEL/Youtube /TEDx/ MIT OW/edX)		NPTEL videos on Testing topic
		14- Lecture Capture Usage		
		15- Any other activity		
4	Tests & Assessments	16- Class Tests/ Weekly Tests		
		17- Pop Quiz		Pop Quiz designed in MS-Form
		18- Mobile App Based Quiz		
		19- Open Book Test		
		20- Take Home Test		on module 4
		21- Any other activity		

No.	Programme	Course	Uploaded on V-refer	Date
1	INFT	SEWDL		

12 Lecture Guide

No.	Programme	Course	Uploaded on V-refer	Date
1	INFT	SEWDL		

*** Do not delete any activity. Give details for planned events. Write 'NA' for activity Not Planned.**

Consolidated Academic Administration Plan Prepared by (mention all theory teaching faculty names with signature)

Please write below your name and sign with date of the external cluster mentor meeting

 **Prof. Deepali Shrikhande**		

External Industry Mentor Ms. Divya Javrani	**External Academic Mentor Dr.Shalu Chopra**	 **Prof. Deepali Shrikhande VIT Cluster Mentor**	**Program HOD**

Annexure:

Assessment Criteria of Learning Outcomes:

Learning Outcomes: The Learner will:	**Assessment Criteria: The Learner can:**	**Evaluated under ISA/MSE/ESE/LAB**
LO1 : A learner will be able to design and develop styled, interactive, and modular web pages using HTML, CSS, and JavaScript.	1.1 Recall and apply fundamental HTML tags to construct static web pages.	LAB / ISA
LO1 : A learner will be able to design and develop styled, interactive, and modular web pages using HTML, CSS, and JavaScript.	1.2 Implement CSS using internal, external, and grid layouts to style web content effectively.	LAB / ISA
LO1 : A learner will be able to design and develop styled, interactive, and modular web pages using HTML, CSS, and JavaScript.	1.3 Use JavaScript to create interactive functionality through DOM manipulation.	LAB / ISA
LO1 : A learner will be able to design and develop styled, interactive, and modular web pages using HTML, CSS, and JavaScript.	1.4Developmodularwebpages integrating HTML, CSS, and JavaScript.	LAB / ISA
LO1 : A learner will be able to design and develop styled, interactive, and modular web pages using HTML, CSS, and JavaScript.	1.5 Debug and resolve issues in front-end code implementations.	LAB
LO2 : A learner will be able to understand, analyse, evaluate, and apply various software process models and methodologies for effective project planning and development.	2.1 Define various software process models including prescriptive and agile models.	MSE / ESE
LO2 : A learner will be able to understand, analyse, evaluate, and apply various software process models and methodologies for effective project planning and development.	2.2 Analyse the suitability of different process models for specific software project scenarios.	ISA / ESE
LO2 : A learner will be able to understand, analyse, evaluate, and apply various software process models and methodologies for effective project planning and development.	2.3 Evaluate pros and cons of iterative and incremental methodologies.	ESE / MSE
LO3 : A learner will be able to define, gather, document, analyse, and refine software requirements using appropriate techniques to create clear and actionable specifications.	3.1 Identify and define functional and non- functional requirements with examples.	ISA / MSE
LO3 : A learner will be able to define, gather, document, analyse, and refine software requirements using appropriate techniques to create clear and actionable specifications.	3.2Gatherrequirements through techniques like interviews, surveys or brainstorming.	ISA
LO3 : A learner will be able to define, gather, document, analyse, and refine software requirements using appropriate techniques to create clear and actionable specifications.	3.3 Document requirements clearly in a SoftwareRequirementsSpecification (SRS).	LAB / ISA
LO4 : A learner will be able to estimate software projects using models like LOC, FP, and COCOMO; define tasks with WBS; develop and track schedules with charts; and analyse project metrics for progress and completion prediction.	4.1 Explain and apply software estimation models such as LOC, FP and COCOMO.	ISA / ESE
LO4 : A learner will be able to estimate software projects using models like LOC, FP, and COCOMO; define tasks with WBS; develop and track schedules with charts; and analyse project metrics for progress and completion prediction.	4.2 Create Work Breakdown Structures (WBS) for software project planning.	LAB / MSE
LO4 : A learner will be able to estimate software projects using models like LOC, FP, and COCOMO; define tasks with WBS; develop and track schedules with charts; and analyse project metrics for progress and completion prediction.	4.3 Develop project schedules using Gantt charts or similar tools.	LAB / MSE
LO5 : A learner will be able to understand core software design principles, explain interaction design fundamentals for usability, and apply design heuristics and	5.1 Define and explain key software design principles and their importance.	ISA / MSE
LO5 : A learner will be able to understand core software design principles, explain interaction design fundamentals for usability, and apply design heuristics and	5.2Describethefundamentalsof interaction design and its role in usability.	ISA